USERS

7 colunas | PK (id)

CHAVE	COLUNA	TIPO	NULO
PK	id DEFAULT IDENTITY	INTEGER	NN
	username UNIQUE	VARCHAR(100)	NN
	email UNIQUE	VARCHAR(255)	NN
	password_hash	VARCHAR(255)	NN
	is_active DEFAULT TRUE	BOOLEAN	NN
	created_at DEFAULT NOW()	TIMESTAMPTZ	NN
	updated_at DEFAULT NOW()	TIMESTAMPTZ	NN

ROLES

5 colunas | PK (id)

CHAVE	COLUNA	TIPO	NULO
PK	id DEFAULT IDENTITY	INTEGER	NN
	name UNIQUE	VARCHAR(50)	NN
	description	TEXT	NULL
	is_system DEFAULT FALSE	BOOLEAN	NN
	created_at DEFAULT NOW()	TIMESTAMPTZ	NN

PERMISSIONS

5 colunas | PK (id)

CHAVE	COLUNA	TIPO	NULO
PK	id DEFAULT IDENTITY	INTEGER	NN
	code UNIQUE	VARCHAR(100)	NN
	description	TEXT	NULL
	module	VARCHAR(100)	NN
	created_at DEFAULT NOW()	TIMESTAMPTZ	NN

USER_ROLES

4 colunas | PK (user_id, role_id)

CHAVE	COLUNA	TIPO	NULO
PK/FK	user_id FK: users.id | DELETE CASCADE	INTEGER	NN
PK/FK	role_id FK: roles.id | DELETE CASCADE	INTEGER	NN
FK	assigned_by FK: users.id | DELETE SET NULL	INTEGER	NULL
	assigned_at DEFAULT NOW()	TIMESTAMPTZ	NN

ROLE_PERMISSIONS

3 colunas | PK (role_id, permission_id)

CHAVE	COLUNA	TIPO	NULO
PK/FK	role_id FK: roles.id | DELETE CASCADE	INTEGER	NN
PK/FK	permission_id FK: permissions.id | DELETE CASCADE	INTEGER	NN
	granted_at DEFAULT NOW()	TIMESTAMPTZ	NN

1 : 0..N

0..1 : 0..N

1 : 0..N

1 : 0..N

1 : 0..N

USERS

PK id : INTEGER

Referência · definida em outra folha

ROLES

PK id : INTEGER

Referência · definida em outra folha

PERMISSIONS

PK id : INTEGER

Referência · definida em outra folha

PK = chave primária · FK = chave estrangeira · NN = NOT NULL · NULL = opcional · UK = UNIQUE composto · Traços: pai : filhos (1 ou 0..1 : 0..N). Caixas pontilhadas repetem somente a chave de outra folha. Exportação local de apoio; salvamento no brModelo Web pendente.

ASSETS

8 colunas | PK (id)

CHAVE	COLUNA	TIPO	NULO
PK	id DEFAULT IDENTITY	INTEGER	NN
	symbol UNIQUE	VARCHAR(20)	NN
	name	VARCHAR(200)	NN
	market	VARCHAR(50)	NN
	asset_type	VARCHAR(50)	NN
	is_active DEFAULT TRUE	BOOLEAN	NN
	created_at DEFAULT NOW()	TIMESTAMPTZ	NN
	updated_at DEFAULT NOW()	TIMESTAMPTZ	NN

ASSET_CONTRACT_SPECS

10 colunas | PK (id)

CHAVE	COLUNA	TIPO	NULO
PK	id DEFAULT IDENTITY	INTEGER	NN
FK	asset_id FK: assets.id | DELETE RESTRICT	INTEGER	NN
	contract_code UNIQUE	VARCHAR(20)	NN
	expiry_date	DATE	NULL
	tick_size	NUMERIC(10,6)	NN
	point_value	NUMERIC(12,4)	NN
	min_lot DEFAULT 1	NUMERIC(10,4)	NN
	lot_step DEFAULT 1	NUMERIC(10,4)	NN
	margin_per_contract DEFAULT 0	NUMERIC(12,2)	NN
	currency DEFAULT 'BRL'	VARCHAR(10)	NN

```
CHECK (tick_size > 0)
CHECK (point_value > 0)
CHECK (min_lot > 0)
CHECK (lot_step > 0)
CHECK (margin_per_contract >= 0)
```

1 : 0..N

PK = chave primária · FK = chave estrangeira · NN = NOT NULL · NULL = opcional · UK = UNIQUE composto · Traços: pai : filhos (1 ou 0..1 : 0..N). Caixas pontilhadas repetem somente a chave de outra folha. Exportação local de apoio; salvamento no brModelo Web pendente.

DATA_PROVIDERS

6 colunas | PK (id)

CHAVE	COLUNA	TIPO	NULO
PK	id DEFAULT IDENTITY	INTEGER	NN
	name UNIQUE	VARCHAR(100)	NN
	type	VARCHAR(50)	NN
	base_url	VARCHAR(500)	NULL
	is_active DEFAULT TRUE	BOOLEAN	NN
	created_at DEFAULT NOW()	TIMESTAMPTZ	NN

TIMEFRAMES

3 colunas | PK (id)

CHAVE	COLUNA	TIPO	NULO
PK	id DEFAULT IDENTITY	INTEGER	NN
	code UNIQUE	VARCHAR(10)	NN
	duration_seconds UNIQUE	INTEGER	NN

CHECK (duration_seconds > 0)

MARKET_DATA

12 colunas | PK (id)

CHAVE	COLUNA	TIPO	NULO
PK	id DEFAULT IDENTITY	BIGINT	NN
FK	asset_id FK: assets.id | DELETE RESTRICT	INTEGER	NN
FK	provider_id FK: data_providers.id | DELETE RESTRICT	INTEGER	NN
FK	timeframe_id FK: timeframes.id | DELETE RESTRICT	INTEGER	NN
	timestamp	TIMESTAMPTZ	NN
	open	NUMERIC(12,4)	NN
	high	NUMERIC(12,4)	NN
	low	NUMERIC(12,4)	NN
	close	NUMERIC(12,4)	NN
	volume DEFAULT 0	NUMERIC(18,4)	NN
	received_at DEFAULT NOW()	TIMESTAMPTZ	NN
	is_complete DEFAULT TRUE	BOOLEAN	NN

UK1 (asset_id, provider_id, timeframe_id, timestamp)
CHECK (volume >= 0)
CHECK (low <= open AND open <= high)
CHECK (low <= close AND close <= high)

1 : 0..N

1 : 0..N

1 : 0..N

ASSETS

PK id : INTEGER

Referência · definida em outra folha

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STRATEGIES

8 colunas | PK (id)

CHAVE	COLUNA	TIPO	NULO
PK	id DEFAULT IDENTITY	INTEGER	NN
	name UNIQUE	VARCHAR(100)	NN
	description	TEXT	NULL
	strategy_type	VARCHAR(20)	NN
	status DEFAULT 'draft'	VARCHAR(20)	NN
	is_enabled DEFAULT FALSE	BOOLEAN	NN
	created_at DEFAULT NOW()	TIMESTAMPTZ	NN
	updated_at DEFAULT NOW()	TIMESTAMPTZ	NN

STRATEGY_VERSIONS

8 colunas | PK (id)

CHAVE	COLUNA	TIPO	NULO
PK	id DEFAULT IDENTITY	INTEGER	NN
FK	strategy_id FK: strategies.id | DELETE RESTRICT	INTEGER	NN
	version	VARCHAR(50)	NN
	changelog	TEXT	NULL
	code_hash	VARCHAR(64)	NULL
	is_active DEFAULT FALSE	BOOLEAN	NN
	released_at	TIMESTAMPTZ	NULL
	created_at DEFAULT NOW()	TIMESTAMPTZ	NN

UK1 (strategy_id, version)

STRATEGY_ASSETS

2 colunas | PK (strategy_version_id, asset_id)

CHAVE	COLUNA	TIPO	NULO
PK/FK	strategy_version_id FK: strategy_versions.id | DELETE CASCADE	INTEGER	NN
PK/FK	asset_id FK: assets.id | DELETE RESTRICT	INTEGER	NN

1 : 0..N

1 : 0..N

1 : 0..N

ASSETS

PK id : INTEGER

Referência · definida em outra folha

PK = chave primária · FK = chave estrangeira · NN = NOT NULL · NULL = opcional · UK = UNIQUE composto · Traços: pai : filhos (1 ou 0..1 : 0..N). Caixas pontilhadas repetem somente a chave de outra folha. Exportação local de apoio; salvamento no brModelo Web pendente.

STRATEGY_TIMEFRAMES

2 colunas | PK (strategy_version_id, timeframe_id)

CHAVE	COLUNA	TIPO	NULO
PK/FK	strategy_version_id FK: strategy_versions.id | DELETE CASCADE	INTEGER	NN
PK/FK	timeframe_id FK: timeframes.id | DELETE RESTRICT	INTEGER	NN

STRATEGY_RULES

6 colunas | PK (id)

CHAVE	COLUNA	TIPO	NULO
PK	id DEFAULT IDENTITY	INTEGER	NN
FK	strategy_version_id FK: strategy_versions.id | DELETE CASCADE	INTEGER	NN
	rule_kind	VARCHAR(10)	NN
	sequence	INTEGER	NN
	expression	TEXT	NN
	description	TEXT	NULL

UK1 (strategy_version_id, rule_kind, sequence)
CHECK (rule_kind IN ('ENTRY', 'EXIT', 'FILTER'))
CHECK (sequence > 0)

STRATEGY_PARAMETERS

7 colunas | PK (id)

CHAVE	COLUNA	TIPO	NULO
PK	id DEFAULT IDENTITY	INTEGER	NN
FK	strategy_version_id FK: strategy_versions.id | DELETE CASCADE	INTEGER	NN
	param_name	VARCHAR(100)	NN
	param_value	TEXT	NN
	param_type DEFAULT 'string'	VARCHAR(20)	NN
	description	TEXT	NULL
	created_at DEFAULT NOW()	TIMESTAMPTZ	NN

UK1 (strategy_version_id, param_name)
CHECK (param_type IN ('string', 'integer', 'decimal', 'boolean'))

1 : 0..N

1 : 0..N

1 : 0..N

1 : 0..N

STRATEGY_VERSIONS

PK id : INTEGER

Referência · definida em outra folha

TIMEFRAMES

PK id : INTEGER

Referência · definida em outra folha

PK = chave primária · FK = chave estrangeira · NN = NOT NULL · NULL = opcional · UK = UNIQUE composto · Traços: pai : filhos (1 ou 0..1 : 0..N). Caixas pontilhadas repetem somente a chave de outra folha. Exportação local de apoio; salvamento no brModelo Web pendente.

AI_MODELS

8 colunas | PK (id)

CHAVE	COLUNA	TIPO	NULO
PK	id DEFAULT IDENTITY	INTEGER	NN
	name UNIQUE	VARCHAR(100)	NN
	description	TEXT	NULL
	model_type	VARCHAR(50)	NN
	framework	VARCHAR(50)	NN
	status DEFAULT 'draft'	VARCHAR(20)	NN
	created_at DEFAULT NOW()	TIMESTAMPTZ	NN
	updated_at DEFAULT NOW()	TIMESTAMPTZ	NN

DATASETS

10 colunas | PK (id)

CHAVE	COLUNA	TIPO	NULO
PK	id DEFAULT IDENTITY	INTEGER	NN
	name	VARCHAR(100)	NN
	description	TEXT	NULL
FK	provider_id FK: data_providers.id | DELETE RESTRICT	INTEGER	NN
FK	asset_id FK: assets.id | DELETE RESTRICT	INTEGER	NN
FK	timeframe_id FK: timeframes.id | DELETE RESTRICT	INTEGER	NN
	start_date	TIMESTAMPTZ	NN
	end_date	TIMESTAMPTZ	NN
	record_count DEFAULT 0	INTEGER	NN
	created_at DEFAULT NOW()	TIMESTAMPTZ	NN

CHECK (end_date >= start_date)
CHECK (record_count >= 0)

MODEL_VERSIONS

9 colunas | PK (id)

CHAVE	COLUNA	TIPO	NULO
PK	id DEFAULT IDENTITY	INTEGER	NN
FK	ai_model_id FK: ai_models.id | DELETE RESTRICT	INTEGER	NN
	version	VARCHAR(50)	NN
	artifact_path	VARCHAR(500)	NULL
	training_started_at	TIMESTAMPTZ	NULL
	training_finished_at	TIMESTAMPTZ	NULL
FK	dataset_id FK: datasets.id | DELETE RESTRICT	INTEGER	NULL
	is_active DEFAULT FALSE	BOOLEAN	NN
	created_at DEFAULT NOW()	TIMESTAMPTZ	NN

UK1 (ai_model_id, version)
CHECK (training_finished_at IS NULL OR (training_started_at IS NOT NULL
AND training_finished_at >= training_started_at))
CHECK (training_started_at IS NULL OR dataset_id IS NOT NULL)

1 : 0..N

1 : 0..N

1 : 0..N

1 : 0..N

0..1 : 0..N

DATA_PROVIDERS

PK id : INTEGER

Referência · definida em outra folha

ASSETS

PK id : INTEGER

Referência · definida em outra folha

TIMEFRAMES

PK id : INTEGER

Referência · definida em outra folha

PK = chave primária · FK = chave estrangeira · NN = NOT NULL · NULL = opcional · UK = UNIQUE composto · Traços: pai : filhos (1 ou 0..1 : 0..N). Caixas pontilhadas repetem somente a chave de outra folha. Exportação local de apoio; salvamento no brModelo Web pendente.

MODEL_METRICS

4 colunas | PK (model_version_id, metric_name)

CHAVE	COLUNA	TIPO	NULO
PK/FK	model_version_id FK: model_versions.id | DELETE CASCADE	INTEGER	NN
PK	metric_name	VARCHAR(100)	NN
	metric_value	NUMERIC(18,6)	NN
	calculated_at DEFAULT NOW()	TIMESTAMPTZ	NN

1 : 0..N

MODEL_VERSIONS

PK id : INTEGER

Referência · definida em outra folha

RISK_PROFILES

16 colunas | PK (id)

CHAVE	COLUNA	TIPO	NULO
PK	id DEFAULT IDENTITY	INTEGER	NN
	name UNIQUE	VARCHAR(100)	NN
	description	TEXT	NULL
	max_daily_loss	NUMERIC(12,2)	NULL
	max_drawdown	NUMERIC(12,2)	NULL
	max_position_size	NUMERIC(12,2)	NULL
	max_exposure_per_asset	NUMERIC(12,2)	NULL
	max_exposure_per_account	NUMERIC(12,2)	NULL
	max_positions	INTEGER	NULL
	max_orders_per_day	INTEGER	NULL
	max_contracts	NUMERIC(12,4)	NULL
	stop_loss_required DEFAULT TRUE	BOOLEAN	NN
	max_slippage	NUMERIC(12,4)	NULL
	max_spread	NUMERIC(12,4)	NULL
	is_active DEFAULT TRUE	BOOLEAN	NN
	created_at DEFAULT NOW()	TIMESTAMPTZ	NN

```
CHECK (max_daily_loss >= 0)
CHECK (max_drawdown >= 0)
CHECK (max_position_size >= 0)
CHECK (max_exposure_per_asset >= 0)
CHECK (max_exposure_per_account >= 0)
CHECK (max_positions >= 0)
CHECK (max_orders_per_day >= 0)
CHECK (max_contracts >= 0)
CHECK (max_slippage >= 0)
CHECK (max_spread >= 0)
```

RISK_EVENTS

11 colunas | PK (id)

CHAVE	COLUNA	TIPO	NULO
PK	id DEFAULT IDENTITY	BIGINT	NN
	event_id UNIQUE	VARCHAR(100)	NN
FK	account_id FK: accounts.id | DELETE RESTRICT	INTEGER	NULL
FK	risk_profile_id FK: risk_profiles.id | DELETE RESTRICT	INTEGER	NULL
FK	asset_id FK: assets.id | DELETE RESTRICT	INTEGER	NULL
FK	strategy_version_id FK: strategy_versions.id | DELETE RESTRICT	INTEGER	NULL
	rule_type	VARCHAR(50)	NN
	decision	VARCHAR(20)	NN
	message	TEXT	NULL
	metadata	TEXT	NULL
	created_at DEFAULT NOW()	TIMESTAMPTZ	NN

0..1 : 0..N

0..1 : 0..N

0..1 : 0..N

0..1 : 0..N

ACCOUNTS

PK id : INTEGER

Referência · definida em outra folha

ASSETS

PK id : INTEGER

Referência · definida em outra folha

STRATEGY_VERSIONS

PK id : INTEGER

Referência · definida em outra folha

ACCOUNTS

11 colunas | PK (id)

CHAVE	COLUNA	TIPO	NULO
PK	id DEFAULT IDENTITY	INTEGER	NN
	name	VARCHAR(100)	NN
	account_number UNIQUE	VARCHAR(100)	NN
FK	risk_profile_id FK: risk_profiles.id | DELETE RESTRICT	INTEGER	NN
	environment	VARCHAR(20)	NN
	currency DEFAULT 'BRL'	VARCHAR(10)	NN
	balance DEFAULT 0	NUMERIC(12,2)	NN
	used_margin DEFAULT 0	NUMERIC(12,2)	NN
	is_active DEFAULT TRUE	BOOLEAN	NN
	created_at DEFAULT NOW()	TIMESTAMPTZ	NN
	updated_at DEFAULT NOW()	TIMESTAMPTZ	NN

```
CHECK (environment IN ('LIVE', 'DEMO', 'PAPER'))
CHECK (used_margin >= 0)
```

BROKER_CONNECTIONS

10 colunas | PK (id)

CHAVE	COLUNA	TIPO	NULO
PK	id DEFAULT IDENTITY	INTEGER	NN
FK	account_id FK: accounts.id | DELETE RESTRICT	INTEGER	NN
	broker_name	VARCHAR(100)	NN
	adapter_name	VARCHAR(100)	NN
	api_key_encrypted	TEXT	NULL
	secret_encrypted	TEXT	NULL
	is_authorized DEFAULT FALSE	BOOLEAN	NN
	connected_at	TIMESTAMPTZ	NULL
	last_heartbeat	TIMESTAMPTZ	NULL
	created_at DEFAULT NOW()	TIMESTAMPTZ	NN

PAPER_SESSIONS

8 colunas | PK (id)

CHAVE	COLUNA	TIPO	NULO
PK	id DEFAULT IDENTITY	BIGINT	NN
FK	account_id FK: accounts.id | DELETE RESTRICT	INTEGER	NN
	name	VARCHAR(100)	NN
	initial_balance	NUMERIC(12,2)	NN
	is_active DEFAULT TRUE	BOOLEAN	NN
	started_at DEFAULT NOW()	TIMESTAMPTZ	NN
	ended_at	TIMESTAMPTZ	NULL
	created_at DEFAULT NOW()	TIMESTAMPTZ	NN

```
CHECK (initial_balance >= 0)
CHECK (ended_at IS NULL OR ended_at >= started_at)
CHECK (ended_at IS NULL OR is_active = FALSE)
```

1 : 0..N

1 : 0..N

1 : 0..N

RISK_PROFILES

PK id : INTEGER

Referência · definida em outra folha

TRADING_SIGNALS

14 colunas | PK (id)

CHAVE	COLUNA	TIPO	NULO
PK	id DEFAULT IDENTITY	BIGINT	NN
FK	strategy_version_id FK: strategy_versions.id | DELETE RESTRICT	INTEGER	NN
FK	asset_id FK: assets.id | DELETE RESTRICT	INTEGER	NN
FK	timeframe_id FK: timeframes.id | DELETE RESTRICT	INTEGER	NN
	side	VARCHAR(4)	NN
	entry_price	NUMERIC(12,4)	NN
	stop_loss	NUMERIC(12,4)	NN
	take_profit	NUMERIC(12,4)	NULL
	quantity	NUMERIC(12,4)	NULL
	justification	TEXT	NULL
	timestamp	TIMESTAMPTZ	NN
	created_at DEFAULT NOW()	TIMESTAMPTZ	NN
	confidence	NUMERIC(3,2)	NULL
	correlation_id DEFAULT gen_random_uuid()	UUID	NN

```
CHECK (side IN ('BUY', 'SELL'))
CHECK (quantity > 0)
CHECK (confidence BETWEEN 0 AND 1)
```

SIGNAL_CONDITIONS

4 colunas | PK (signal_id, rule_id)

CHAVE	COLUNA	TIPO	NULO
PK/FK	signal_id FK: trading_signals.id | DELETE CASCADE	BIGINT	NN
PK/FK	rule_id FK: strategy_rules.id | DELETE RESTRICT	INTEGER	NN
	is_met	BOOLEAN	NN
	observed_value	TEXT	NULL

APPROVALS

8 colunas | PK (id)

CHAVE	COLUNA	TIPO	NULO
PK	id DEFAULT IDENTITY	BIGINT	NN
FK	signal_id FK: trading_signals.id | DELETE RESTRICT	BIGINT	NN
FK	user_id FK: users.id | DELETE RESTRICT	INTEGER	NN
	decision	VARCHAR(10)	NN
	approved_at	TIMESTAMPTZ	NULL
	expires_at	TIMESTAMPTZ	NULL
	justification	TEXT	NULL
	created_at DEFAULT NOW()	TIMESTAMPTZ	NN

```
CHECK (decision IN ('APPROVED', 'REJECTED'))
CHECK ((decision = 'APPROVED' AND approved_at IS NOT NULL) OR (decision
= 'REJECTED' AND approved_at IS NULL))
CHECK (expires_at IS NULL OR (approved_at IS NOT NULL AND expires_at >
approved_at))
```

1 : 0..N

1 : 0..N

1 : 0..N

1 : 0..N

1 : 0..N

1 : 0..N

1 : 0..N

STRATEGY_VERSIONS
PK id : INTEGER
Referência · definida em outra folha

ASSETS
PK id : INTEGER
Referência · definida em outra folha

TIMEFRAMES
PK id : INTEGER
Referência · definida em outra folha

STRATEGY_RULES
PK id : INTEGER
Referência · definida em outra folha

USERS
PK id : INTEGER
Referência · definida em outra folha

PK = chave primária · FK = chave estrangeira · NN = NOT NULL · NULL = opcional · UK = UNIQUE composto · Traços: pai : filhos (1 ou 0..1 : 0..N). Caixas pontilhadas repetem somente a chave de outra folha. Exportação local de apoio; salvamento no brModelo Web pendente.

ORDERS

17 colunas | PK (id)

CHAVE	COLUNA	TIPO	NULO
PK	id DEFAULT IDENTITY	BIGINT	NN
	order_id UNIQUE	VARCHAR(100)	NN
	client_order_id UNIQUE	VARCHAR(100)	NULL
	broker_order_id	VARCHAR(100)	NULL
FK	account_id FK: accounts.id | DELETE RESTRICT	INTEGER	NN
FK	asset_id FK: assets.id | DELETE RESTRICT	INTEGER	NN
FK	signal_id FK: trading_signals.id | DELETE RESTRICT	BIGINT	NULL
	side	VARCHAR(4)	NN
	order_type	VARCHAR(12)	NN
	quantity	NUMERIC(12,4)	NN
	price	NUMERIC(12,4)	NULL
	stop_price	NUMERIC(12,4)	NULL
	status DEFAULT 'INTENTION'	VARCHAR(20)	NN
	rejection_reason	VARCHAR(500)	NULL
	correlation_id DEFAULT gen_random_uuid()	UUID	NN
	created_at DEFAULT NOW()	TIMESTAMPTZ	NN
	updated_at DEFAULT NOW()	TIMESTAMPTZ	NN

```
UK1 (account_id, broker_order_id)
CHECK (side IN ('BUY', 'SELL'))
CHECK (quantity > 0)
CHECK (order_type IN ('MARKET', 'LIMIT', 'STOP', 'STOP_LIMIT'))
CHECK (order_type NOT IN ('LIMIT', 'STOP_LIMIT') OR price IS NOT NULL)
CHECK (order_type NOT IN ('STOP', 'STOP_LIMIT') OR stop_price IS NOT
NULL)
```

ORDER_EXECUTIONS

7 colunas | PK (id)

CHAVE	COLUNA	TIPO	NULO
PK	id DEFAULT IDENTITY	BIGINT	NN
	execution_id UNIQUE	VARCHAR(100)	NN
FK	order_id FK: orders.id | DELETE RESTRICT	BIGINT	NN
	quantity	NUMERIC(12,4)	NN
	price	NUMERIC(12,4)	NN
	commission DEFAULT 0	NUMERIC(12,4)	NN
	executed_at DEFAULT NOW()	TIMESTAMPTZ	NN

```
CHECK (quantity > 0)
CHECK (commission >= 0)
```

1 : 0..N

1 : 0..N

0..1 : 0..N

1 : 0..N

ACCOUNTS

PK id : INTEGER

Referência · definida em outra folha

ASSETS

PK id : INTEGER

Referência · definida em outra folha

TRADING_SIGNALS

PK id : BIGINT

Referência · definida em outra folha

PORTFOLIOS

5 colunas | PK (id)

CHAVE	COLUNA	TIPO	NULO
PK	id DEFAULT IDENTITY	INTEGER	NN
FK	account_id FK: accounts.id | DELETE RESTRICT	INTEGER	NN
	name	VARCHAR(100)	NN
	created_at DEFAULT NOW()	TIMESTAMPTZ	NN
	updated_at DEFAULT NOW()	TIMESTAMPTZ	NN

UK1 (account_id, name)

POSITIONS

9 colunas | PK (id)

CHAVE	COLUNA	TIPO	NULO
PK	id DEFAULT IDENTITY	BIGINT	NN
	position_id UNIQUE	VARCHAR(100)	NN
FK	portfolio_id FK: portfolios.id | DELETE RESTRICT	INTEGER	NN
FK	asset_id FK: assets.id | DELETE RESTRICT	INTEGER	NN
	side	VARCHAR(5)	NN
	stop_loss	NUMERIC(12,4)	NULL
	take_profit	NUMERIC(12,4)	NULL
	opened_at DEFAULT NOW()	TIMESTAMPTZ	NN
	closed_at	TIMESTAMPTZ	NULL

CHECK (side IN ('LONG', 'SHORT'))
CHECK (closed_at IS NULL OR closed_at >= opened_at)

PORTFOLIO_SNAPSHOTS

11 colunas | PK (id)

CHAVE	COLUNA	TIPO	NULO
PK	id DEFAULT IDENTITY	BIGINT	NN
FK	portfolio_id FK: portfolios.id | DELETE RESTRICT	INTEGER	NN
	snapshot_id UNIQUE	VARCHAR(100)	NN
	total_capital	NUMERIC(12,2)	NN
	used_margin	NUMERIC(12,2)	NN
	total_exposure	NUMERIC(12,2)	NN
	total_pnl_unrealized	NUMERIC(12,2)	NN
	total_pnl_realized	NUMERIC(12,2)	NN
	total_commission	NUMERIC(12,2)	NN
	open_positions_count	INTEGER	NN
	timestamp DEFAULT NOW()	TIMESTAMPTZ	NN

UK1 (portfolio_id, timestamp)
CHECK (used_margin >= 0)
CHECK (total_exposure >= 0)
CHECK (total_commission >= 0)
CHECK (open_positions_count >= 0)

1 : 0..N

1 : 0..N

1 : 0..N

1 : 0..N

ACCOUNTS

PK id : INTEGER

Referência · definida em outra folha

ASSETS

PK id : INTEGER

Referência · definida em outra folha

PK = chave primária · FK = chave estrangeira · NN = NOT NULL · NULL = opcional · UK = UNIQUE composto · Traços: pai : filhos (1 ou 0..1 : 0..N). Caixas pontilhadas repetem somente a chave de outra folha. Exportação local de apoio; salvamento no brModelo Web pendente.

POSITION_EXECUTIONS

4 colunas | PK (position_id, execution_id)

CHAVE	COLUNA	TIPO	NULO
PK/FK	position_id FK: positions.id | DELETE RESTRICT	BIGINT	NN
PK/FK	execution_id FK: order_executions.id | DELETE RESTRICT	BIGINT	NN
	effect	VARCHAR(5)	NN
	allocated_quantity	NUMERIC(12,4)	NN

```
CHECK (effect IN ('OPEN', 'CLOSE'))
CHECK (allocated_quantity > 0)
```

1 : 0..N

1 : 0..N

POSITIONS

PK id : BIGINT

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ORDER_EXECUTIONS

PK id : BIGINT

Referência · definida em outra folha

AGENT_SESSIONS

9 colunas | PK (id)

CHAVE	COLUNA	TIPO	NULO
PK	id DEFAULT IDENTITY	BIGINT	NN
	session_id UNIQUE	VARCHAR(100)	NN
FK	user_id FK: users.id | DELETE RESTRICT	INTEGER	NN
FK	account_id FK: accounts.id | DELETE RESTRICT	INTEGER	NN
	mode	VARCHAR(20)	NN
	is_active DEFAULT TRUE	BOOLEAN	NN
	started_at DEFAULT NOW()	TIMESTAMPTZ	NN
	ended_at	TIMESTAMPTZ	NULL
	config_snapshot	TEXT	NULL

```
CHECK (ended_at IS NULL OR ended_at >= started_at)
CHECK (ended_at IS NULL OR is_active = FALSE)
```

AUDIT_LOGS

11 colunas | PK (id)

CHAVE	COLUNA	TIPO	NULO
PK	id DEFAULT IDENTITY	BIGINT	NN
	log_id UNIQUE	VARCHAR(100)	NN
FK	user_id FK: users.id | DELETE SET NULL	INTEGER	NULL
	action	VARCHAR(100)	NN
	entity_type	VARCHAR(50)	NN
	entity_id	VARCHAR(100)	NN
	old_value	TEXT	NULL
	new_value	TEXT	NULL
	environment	VARCHAR(20)	NN
	ip_address	VARCHAR(45)	NULL
	created_at DEFAULT NOW()	TIMESTAMPTZ	NN

SYSTEM_EVENTS

10 colunas | PK (id)

CHAVE	COLUNA	TIPO	NULO
PK	id DEFAULT IDENTITY	BIGINT	NN
	event_id UNIQUE	VARCHAR(100)	NN
	event_type	VARCHAR(50)	NN
	severity	VARCHAR(20)	NN
	source	VARCHAR(100)	NN
	message	TEXT	NN
	metadata	TEXT	NULL
	environment	VARCHAR(20)	NN
FK	account_id FK: accounts.id | DELETE SET NULL	INTEGER	NULL
	created_at DEFAULT NOW()	TIMESTAMPTZ	NN

1 : 0..N

1 : 0..N

0..1 : 0..N

0..1 : 0..N

USERS

PK id : INTEGER

Referência · definida em outra folha

ACCOUNTS

PK id : INTEGER

Referência · definida em outra folha